

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
nharihatim@nharihatim:~$ ls /etc/ > file.txt
nharihatim@nharihatim:~$ ls >> file.txt
nharihatim@nharihatim:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

```
nharihatim@nharihatim:~$  
nharihatim@nharihatim:~$ grep .conf file.txt > conf.txt  
nharihatim@nharihatim:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprindd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf  
locale.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/project-personal/stage04/report/_resources/csl  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/project-personal/stage04/report/bib/cite.bib  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/project-personal/stage05/report/_resources/csl  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/project-personal/stage05/report/bib/cite.bib  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/project-personal/stage06/report/_resources/csl  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/project-personal/stage06/report/bib/cite.bib  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/presentation/report/_resources/csl  
/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--  
os-intro/presentation/report/bib/cite.bib  
/home/nharihatim/git-extended/.git/hooks/commit-msg.sample  
/home/nharihatim/git-extended/.git/objects/05/cb65f905a83d2c0301bffe21a94b  
5e0b082129  
/home/nharihatim/git-extended/.git/objects/cb  
/home/nharihatim/git-extended/.git/config  
/home/nharihatim/conf.txt  
nharihatim@nharihatim:~$
```

Рисунок 3: Поиск файлов

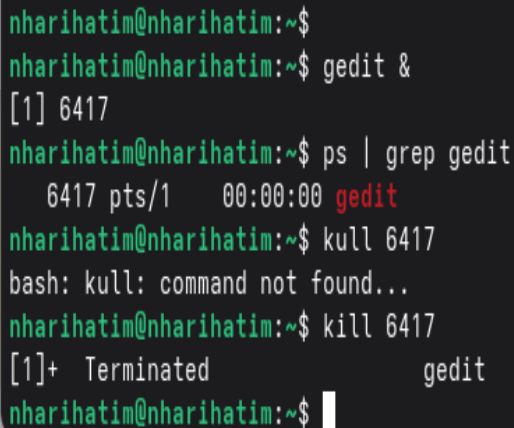
```
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

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Рисунок 4: Поиск файлов

```
nharihatim@nharihatim:~$  
nharihatim@nharihatim:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
find: '/etc/credstore.encrypted': Permission denied  
find: '/etc/credstore': Permission denied  
find: '/etc/cups/ssl': Permission denied  
nharihatim@nharihatim:~$  
nharihatim@nharihatim:~$  
nharihatim@nharihatim:~$ find ~ -name "log*" > logfile &  
[1] 6381  
nharihatim@nharihatim:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
nharihatim@nharihatim:~$ rm logfile  
nharihatim@nharihatim:~$
```

Рисунок 5: Фоновый запуск процесса



```
nharihatim@nharihatim:~$  
nharihatim@nharihatim:~$ gedit &  
[1] 6417  
nharihatim@nharihatim:~$ ps | grep gedit  
    6417 pts/1    00:00:00 gedit  
nharihatim@nharihatim:~$ kull 6417  
bash: kull: command not found...  
nharihatim@nharihatim:~$ kill 6417  
[1]+  Terminated                  gedit  
nharihatim@nharihatim:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df. df displays the
    amount of space available on the file system containing each file
    name argument. If no file name is given, the space available on
    all currently mounted file systems is shown. Space is shown in 1K
    blocks by default, unless the environment variable POSIXLY_CORRECT
    is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node contain-
    ing a mounted file system, df shows the space available on that
    file system rather than on the file system containing the device
    node. This version of df cannot show the space available on un-
    mounted file systems, because on most kinds of systems doing so
    requires non-portable intimate knowledge of file system struc-
    tures.

OPTIONS
    Show information about the file system on which each FILE resides,
    or all file systems by default.

    Mandatory arguments to long options are mandatory for short op-
    tions too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for direc-
    tories.

    Mandatory arguments to long options are mandatory for short op-
    tions too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the
        apparent size is usually smaller, it may be larger due to
        holes in ('sparse') files, internal fragmentation, indirect
        blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM'
        prints sizes in units of 1,048,576 bytes; see SIZE format
        below

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
nharihatim@nharihatim:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91224744 63365224 60% /
devtmpfs        4011284      0    4011284  0% /dev
tmpfs           4035656      92    4035564  1% /dev/shm
tmpfs           1614264     1952    1612312  1% /run
tmpfs           1024         0        1024  0% /run/credentials/systemd-
journald.service
tmpfs           4035660      80    4035580  1% /tmp
/dev/nvme0n1p3 155186176 91224744 63365224 60% /home
/dev/nvme0n1p2 1992552     518800    1352512 28% /boot
/dev/loop1      68480       68480         0 100% /var/lib/snapd/snap/core2
4/1349
/dev/loop2      49280       49280         0 100% /var/lib/snapd/snap/snapd
/25935
/dev/loop0      115712     115712         0 100% /var/lib/snapd/snap/hugo/
25676
tmpfs           1024         0        1024  0% /run/credentials/systemd-
resolved.service
tmpfs           807128      168    806960  1% /run/user/1063
/dev/sr0        2677920    2677920         0 100% /run/media/nharihatim/Fed
ora-WS-Live-43
nharihatim@nharihatim:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/fd
4      ./git-extended/.git/objects/95
4      ./git-extended/.git/objects/d3
4      ./git-extended/.git/objects/cb
4      ./git-extended/.git/objects/9e
4      ./git-extended/.git/objects/11
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
812988 .
nharihatim@nharihatim:~$
```

Рисунок 10: Запуск команды du


```
/home/nharihatim/git-extended/.git/objects/00  
/home/nharihatim/git-extended/.git/objects/ee  
/home/nharihatim/git-extended/.git/objects/d2  
/home/nharihatim/git-extended/.git/objects/77  
/home/nharihatim/git-extended/.git/objects/ac  
/home/nharihatim/git-extended/.git/objects/fd  
/home/nharihatim/git-extended/.git/objects/95  
/home/nharihatim/git-extended/.git/objects/d3  
/home/nharihatim/git-extended/.git/objects/cb  
/home/nharihatim/git-extended/.git/objects/9e  
/home/nharihatim/git-extended/.git/objects/11  
/home/nharihatim/git-extended/.git/logs  
/home/nharihatim/git-extended/.git/logs/refs  
/home/nharihatim/git-extended/.git/logs/refs/heads  
/home/nharihatim/git-extended/.git/logs/refs/remotes  
/home/nharihatim/git-extended/.git/logs/refs/remotes/origin  
/home/nharihatim/monthly  
/home/nharihatim/reports  
/home/nharihatim/reports/monthly  
/home/nharihatim/reports/monthly/monthly  
/home/nharihatim/ski.plases  
/home/nharihatim/ski.plases/equipment  
/home/nharihatim/ski.plases/plans  
/home/nharihatim/australia  
/home/nharihatim/play  
/home/nharihatim/play/games  
/home/nharihatim/play/games/play  
nharihatim@nharihatim:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.